

MYSQL VIEWS

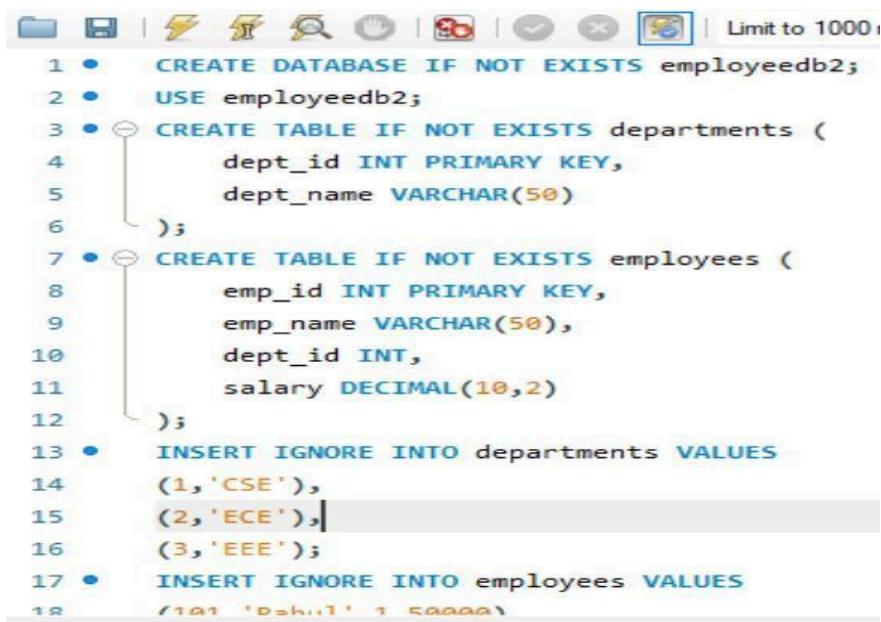
Practical Examples and Student Exercises using MySQL Workbench

1. What is a View?

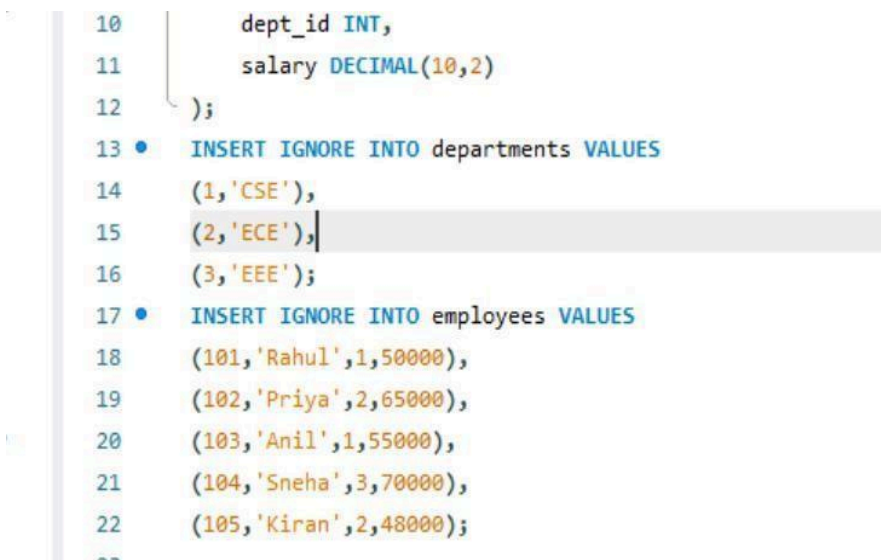
A **View** is a **virtual table** created from one or more existing tables.

- It **does not store data** itself.
- It stores only the **SQL query**.

2. Setup: Employees and Departments Tables



```
1 • CREATE DATABASE IF NOT EXISTS employeeedb2;
2 • USE employeeedb2;
3 • CREATE TABLE IF NOT EXISTS departments (
4     dept_id INT PRIMARY KEY,
5     dept_name VARCHAR(50)
6 );
7 • CREATE TABLE IF NOT EXISTS employees (
8     emp_id INT PRIMARY KEY,
9     emp_name VARCHAR(50),
10    dept_id INT,
11    salary DECIMAL(10,2)
12 );
13 • INSERT IGNORE INTO departments VALUES
14     (1, 'CSE'),
15     (2, 'ECE'),
16     (3, 'EEE');
17 • INSERT IGNORE INTO employees VALUES
18     (101, 'Rahul', 1, 50000);
```



```
10    dept_id INT,
11    salary DECIMAL(10,2)
12 );
13 • INSERT IGNORE INTO departments VALUES
14     (1, 'CSE'),
15     (2, 'ECE'),
16     (3, 'EEE');
17 • INSERT IGNORE INTO employees VALUES
18     (101, 'Rahul', 1, 50000),
19     (102, 'Priya', 2, 65000),
20     (103, 'Anil', 1, 55000),
21     (104, 'Sneha', 3, 70000),
22     (105, 'Kiran', 2, 48000);
```

3. Verify the Tables

The screenshot shows a sequence of SQL commands and their results in SQL Developer. The commands are:

```
22 (105, 'Kiran', 2, 48000);
23
24 • SHOW TABLES;
25 • SELECT * FROM departments;
26 • SELECT * FROM employees;
27
28
```

The first result grid shows the tables in the database:

Tables in employeesdb2
departments
employees

The second result grid shows the data from the departments table:

dept_id	dept_name
1	CSE
2	ECE
3	EEE
NULL	NULL

The third result grid shows the data from the employees table:

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

View 1 – Display All Employees

Question: Create a View named EmployeeView to display all columns from employees.

The screenshot shows the SQL commands to create and query the EmployeeView:

```
27
28 • DROP VIEW IF EXISTS EmployeeView;
29
30 • CREATE VIEW EmployeeView AS
31 SELECT *
32 FROM employees;
33
34 • SELECT * FROM EmployeeView;
```

The result grid displays the data from the EmployeeView:

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

View 2 – Selected Columns

Question: Create a View showing employee ID, employee name and salary only.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
35  
36 DROP VIEW IF EXISTS EmployeeBasicView;  
37 CREATE VIEW EmployeeBasicView AS  
38 SELECT emp_id, emp_name, salary  
39 FROM employees;  
40 SELECT * FROM EmployeeBasicView;  
41  
42
```

The Results pane shows the output of the last query, displaying a table with 5 rows and 3 columns: emp_id, emp_name, and salary.

emp_id	emp_name	salary
101	Rahul	50000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

View 3 – High Salary Employees

Question: Create a View that displays employees earning more than 55000.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
42 DROP VIEW IF EXISTS HighSalaryEmployees;  
43 CREATE VIEW HighSalaryEmployees AS  
44 SELECT emp_id, emp_name, salary  
45 FROM employees  
46 WHERE salary > 55000;  
47 SELECT * FROM HighSalaryEmployees;  
48  
49
```

The Results pane shows the output of the last query, displaying a table with 2 rows and 3 columns: emp_id, emp_name, and salary.

emp_id	emp_name	salary
102	Priya	65000.00
104	Sneha	70000.00

View 4 – Employee and Department using JOIN

Question: Create a View that displays employee name, department name and salary.

SQL Editor:

```

50 • CREATE VIEW EmployeeDepartmentView AS
51 SELECT
52     e.emp_id,
53     e.emp_name,
54     d.dept_name,
55     e.salary
56 FROM employees e
57 JOIN departments d
58 ON e.dept_id = d.dept_id;
59 • SELECT * FROM EmployeeDepartmentView;

```

Result Grid:

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	102	Priya	ECE	65000.00
	103	Anil	CSE	55000.00
	104	Sneha	EEE	70000.00
	105	Kiran	ECE	48000.00

EmployeeDepartmentView 7 x

View 5 – Employees from a Particular Department

Question: Create a View that displays only CSE employees.

SQL Editor:

```

61 • DROP VIEW IF EXISTS CSEEmployees;
62 • CREATE VIEW CSEEmployees AS
63 SELECT
64     e.emp_id,
65     e.emp_name,
66     d.dept_name,
67     e.salary
68 FROM employees e
69 JOIN departments d
70 ON e.dept_id = d.dept_id

```

Result Grid:

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	103	Anil	CSE	55000.00

SQL Editor:

```

68 FROM employees e
69 JOIN departments d
70 ON e.dept_id = d.dept_id
71 WHERE d.dept_name = 'CSE';
72 • SELECT * FROM CSEEmployees;

```

Result Grid:

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	103	Anil	CSE	55000.00

View 6 – Department-wise Average Salary

Question: Create a View using GROUP BY and AVG() to display department-wise average salary.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
75 • CREATE VIEW DepartmentSalaryView AS
76   SELECT
77     d.dept_name,
78     COUNT(e.emp_id) AS EmployeeCount,
79     AVG(e.salary) AS AverageSalary
80   FROM employees e
81   JOIN departments d
82   ON e.dept_id = d.dept_id
83   GROUP BY d.dept_id, d.dept_name;
84 • SELECT * FROM DepartmentSalaryView;
```

Below the editor, the 'Result Grid' tab is active, displaying the following data:

dept_name	EmployeeCount	AverageSalary
CSE	2	52500.000000
ECE	2	56500.000000
EEE	1	70000.000000

View 7 – Departments with Average Salary Above 55000

Question: Create a View that displays departments whose average salary is greater than 55000.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
87 • CREATE VIEW HighAverageDepartments AS
88   SELECT
89     d.dept_name,
90     AVG(e.salary) AS AverageSalary
91   FROM employees e
92   JOIN departments d
93   ON e.dept_id = d.dept_id
94   GROUP BY d.dept_id, d.dept_name
95   HAVING AVG(e.salary) > 55000;
96 • SELECT * FROM HighAverageDepartments;
```

Below the editor, the 'Result Grid' tab is active, displaying the following data:

dept_name	AverageSalary
ECE	56500.000000
EEE	70000.000000

View 8 – Salary Range View

Question: Create a View that displays employees with salary between 50000 and 70000.


```

97
98 • DROP VIEW IF EXISTS SalaryRangeView;
99
100 • CREATE VIEW SalaryRangeView AS
101 SELECT emp_id, emp_name, salary
102 FROM employees
103 WHERE salary BETWEEN 50000 AND 70000;
104
105 • SELECT * FROM SalaryRangeView;
106

```

emp_id	emp_name	salary
101	Rahul	50000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00

View 9 – View with Calculated Column

Question: Create a View that displays employee name, monthly salary and annual salary.

```

107 • DROP VIEW IF EXISTS AnnualSalaryView;
108 • CREATE VIEW AnnualSalaryView AS
109 SELECT
110     emp_id,
111     emp_name,
112     salary AS MonthlySalary,
113     salary * 12 AS AnnualSalary
114 FROM employees;
115 • SELECT * FROM AnnualSalaryView;
116

```

emp_id	emp_name	MonthlySalary	AnnualSalary
101	Rahul	50000.00	600000.00
102	Priya	65000.00	780000.00
103	Anil	55000.00	660000.00
104	Sneha	70000.00	840000.00
105	Kiran	48000.00	576000.00

View 10 – Update Data Through a Simple View

Question: Create a simple single-table View and update an employee's salary through the View.

```

116
117 • DROP VIEW IF EXISTS EmployeeSalaryView;
118 • CREATE VIEW EmployeeSalaryView AS
119     SELECT emp_id, emp_name, salary
120     FROM employees;
121 • UPDATE EmployeeSalaryView
122     SET salary = 60000
123     WHERE emp_id = 101;
124 • SELECT * FROM EmployeeSalaryView;
125 • SELECT * FROM employees WHERE emp_id = 101;

```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	60000.00
NULL	NULL	NULL	NULL

View 11 – CREATE OR REPLACE VIEW

Question: Modify an existing View so that it displays employee name and salary only.

```

127 • CREATE OR REPLACE VIEW EmployeeBasicView AS
128     SELECT emp_name, salary
129     FROM employees;
130
131 • SELECT * FROM EmployeeBasicView;
132
133
134

```

Result Grid

emp_name	salary
Rahul	60000.00
Priya	65000.00
Anil	55000.00
Sneha	70000.00
Kiran	48000.00

View 12 – View Definition

Question: Display the SQL statement used to create a View.

```

132
133 • SHOW CREATE VIEW EmployeeDepartmentView;
134

```

Result Grid				
	Filter Rows:		Export:	Wrap Cell Content:
View	Create View	character_set_client	collation_connection	
▶ employeedepartmentview	CREATE ALGORITHM=UNDEFINED DEFINER='r...	utf8mb4	utf8mb4_0900_ai_ci	

View 13 – List All Views

Question: Display all Views available in the current database.

```

134
135 • SHOW FULL TABLES
136 WHERE Table_type = 'VIEW';
137
138

```

Result Grid		
	Filter Rows:	Export: Wrap
Tables_in_employeedb2	Table_type	
▶ annualsalaryview	VIEW	
cseemployees	VIEW	
departmentsalaryview	VIEW	
employeebasicview	VIEW	
employeedepartmentview	VIEW	

Result 17 x

View 14 – Drop a View

Question: Remove an existing View from the database.

```

138 • DROP VIEW IF EXISTS SalaryRangeView;

```

Output				
Action Output				
#	Time	Action	Message	
✓ 40	10:19:17	SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000	1 row(s) returned	
✓ 41	10:22:41	CREATE OR REPLACE VIEW EmployeeBasicView AS SELECT emp_name, salary...	0 row(s) affected	
✓ 42	10:22:41	SELECT * FROM EmployeeBasicView LIMIT 0, 1000	5 row(s) returned	
✓ 43	10:23:27	SHOW CREATE VIEW EmployeeDepartmentView	1 row(s) returned	
✓ 44	10:24:05	SHOW FULL TABLES WHERE Table_type = 'VIEW'	10 row(s) returned	
✓ 45	10:24:38	DROP VIEW IF EXISTS SalaryRangeView	0 row(s) affected	

6. Student Practice Tasks

1. Create a View showing only emp_id, emp_name and salary.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
35  
36 DROP VIEW IF EXISTS EmployeeBasicView;  
37 CREATE VIEW EmployeeBasicView AS  
38 SELECT emp_id, emp_name, salary  
39 FROM employees;  
40 SELECT * FROM EmployeeBasicView;  
41  
42
```

Below the editor, the 'Result Grid' tab is active, displaying the following data:

emp_id	emp_name	salary
101	Rahul	50000.00
102	Priya	65000.00
103	Anil	55000.00
104	Sneha	70000.00
105	Kiran	48000.00

2. Create a View for employees earning more than 60000.

The screenshot shows the SQL Developer interface. The SQL Editor contains the following code:

```
139  
140 DROP VIEW IF EXISTS HighSalaryEmployee;  
141 CREATE VIEW HighSalaryEmployee AS  
142 SELECT emp_id, emp_name, salary  
143 FROM employees  
144 WHERE salary > 60000;  
145 SELECT * FROM HighSalaryEmployee;
```

Below the editor, the 'Result Grid' tab is active, displaying the following data:

emp_id	emp_name	salary
102	Priya	65000.00
104	Sneha	70000.00

3. Create a View showing employee name, department name and salary using JOIN.

146

```

147 • CREATE VIEW employee_department_view AS
148     SELECT e.emp_name, d.dept_name, e.salary
149     FROM employees e
150     JOIN departments d
151     ON e.dept_id = d.dept_id;
152 • SELECT * FROM employee_department_view;

```

Result Grid | Filter Rows: | Export: | Wra

emp_name	dept_name	salary
Rahul	CSE	60000.00
Priya	ECE	65000.00
Anil	CSE	55000.00
Sneha	EEE	70000.00
Kiran	ECE	48000.00

employee_department_view 19 x

4. Create a View showing only ECE employees.

154 • DROP VIEW IF EXISTS ECEEmployees;

```

155 • CREATE VIEW ECEEmployees AS
156     SELECT
157         e.emp_id, e.emp_name,
158         d.dept_name, e.salary
159     FROM employees e
160     JOIN departments d
161     ON e.dept_id = d.dept_id
162     WHERE d.dept_name = 'ECE';
163 • SELECT * FROM ECEEmployees;

```

Result Grid | Filter Rows: | Export: | Limit to 100

emp_id	emp_name	dept_name	salary
102	Priya	ECE	65000.00
105	Kiran	ECE	48000.00

5. Create a View showing department-wise employee count.

164

165 • **CREATE VIEW** dept_employee_count **AS**

166 **SELECT** d.dept_name, COUNT(e.emp_id) **AS** employee_count

167 **FROM** departments d

168 **LEFT JOIN** employees e

169 **ON** d.dept_id = e.dept_id

170 **GROUP BY** d.dept_name;

171 • **SELECT * FROM** dept_employee_count;

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	dept_name	employee_count
▶	CSE	2
	ECE	2
	EEE	1

6. Create a View showing department-wise maximum salary.

172

173 • **CREATE VIEW** dept_max_salary **AS**

174 **SELECT** d.dept_name, MAX(e.salary) **AS** max_salary

175 **FROM** departments d

176 **JOIN** employees e

177 **ON** d.dept_id = e.dept_id

178 **GROUP BY** d.dept_name;

179 • **SELECT * FROM** dept_max_salary;

Result Grid | | Filter Rows: | Export: | Wrap Cell Conte

	dept_name	max_salary
▶	CSE	60000.00
	ECE	65000.00
	EEE	70000.00

7. Create a View showing departments whose average salary is greater than 50000.

```
180
181 • DROP VIEW IF EXISTS HighSalaryEmployeee;
182 • CREATE VIEW HighSalaryEmployeee AS
183     SELECT emp_id, emp_name, salary
184     FROM employees
185     WHERE salary > 50000;
186 • SELECT * FROM HighSalaryEmployeee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	emp_id	emp_name	salary
▶	101	Rahul	60000.00
	102	Priya	65000.00
	103	Anil	55000.00
	104	Sneha	70000.00

HighSalaryEmployeee 23 x

8. Create a View with employee name, salary and annual salary.

```
188 • DROP VIEW IF EXISTS AnnualSalaryViews;
189 • CREATE VIEW AnnualSalaryViews AS
190     SELECT
191         emp_id,
192         emp_name,
193         salary * 12 AS AnnualSalary
194     FROM employees;
195 • SELECT * FROM AnnualSalaryViews;
```

Result Grid | Filter Rows: | Export: | Wrap C

	emp_id	emp_name	AnnualSalary
▶	101	Rahul	720000.00
	102	Priya	780000.00
	103	Anil	660000.00
	104	Sneha	840000.00
	105	Kiran	576000.00

AnnualSalaryViews 24 x

9. Update an employee's salary through a simple View and verify the base table.

```

197 • CREATE VIEW EmployeeSalary AS
198     SELECT emp_id, emp_name, salary
199     FROM employees;
200 • UPDATE EmployeeSalary
201     SET salary = 65000
202     WHERE emp_id = 101;
203 • SELECT * FROM EmployeeSalary
204     WHERE emp_id = 101;
205 • SELECT * FROM employees
206     WHERE emp_id = 101;

```

Result Grid

	emp_id	emp_name	salary
▶	101	Rahul	65000.00

```

202     WHERE emp_id = 101;
203 • SELECT * FROM EmployeeSalary
204     WHERE emp_id = 101;
205 • SELECT * FROM employees
206     WHERE emp_id = 101;

```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	65000.00
*	NULL	NULL	NULL	NULL

10. Use SHOW CREATE VIEW to display the definition of a View.

```

207
208 • SHOW CREATE VIEW AnnualSalaryViews;

```

Result Grid

	View	Create View	character_set_client	collation_connection
▶	annualsalaryviews	CREATE ALGORITHM=UNDEFINED DEFINER=`r...`	utf8mb4	utf8mb4_0900_ai_ci

11. List all Views in the database.

209

210 • `SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW';`

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Tables_in_employeedb2	Table_type
annualsexyview	VIEW
annualsexyviews	VIEW
cseemployees	VIEW
departmentssalaryview	VIEW
dept_employee_count	VIEW

Result 28 x

12. Drop a View and verify that it no longer appears in the list.

211

212 • `DROP VIEW IF EXISTS AnnualSalaryViews;`

213 • `SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW';`

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Tables_in_employeedb2	Table_type
annualsexyview	VIEW
cseemployees	VIEW
departmentssalaryview	VIEW
dept_employee_count	VIEW
dept_max_salary	VIEW

Result 30 x

1.